

Ski & Snowboard Rental Shop Website

Project plan



Date: 12/09/2024

Version: 1.0

Contents

1. INTRODUCTION.....	3
2. Team	3
3. Client.....	3
4. DELIVERABLES	4
5. PHASING	4
6. MILESTONES AND DEADLINES.....	6
7. CONSTRAINTS	7

1. INTRODUCTION

The purpose of this document is to get a clear overview of the project and its planning. This document will include the planning for the project starting in 12 September until 11 October.

Objective: Develop a software system consisting of:

1. A **web application** for users to reserve ski and snowboard equipment.
2. A **desktop application** for managing reservations and rentals in the store.
3. A shared **MSSQL database** connecting both applications, without using ORM or LINQ.

2. Team

My name is Tifrea Andrei (4150104) and I will be responsible for creating the software solution for Tech Initiative. I can be contacted on the following email address 455422@student.fontys.nl.

3. Client

The client for our project is the company Tech Initiative. I will keep in touch with Mr. Stan Hartingsveldt, Koehorst Michiel and Roa Villescas Martin who will represent the company Tech Initiative.

Mr. Stan Hartingsveldt can be reached at the following e-mail address: s.vanhartingsveldt@fontys.nl

Mr. Koehorst Michiel can be reached at the following e-mail address: m.koehorst@fontys.nl

Mr. Roa Villescas Martin can be reached at the following e-mail address: m.roavillescas@fontys.nl

4. DELIVERABLES

1. **Ideation document:** Submitted in Week 1.
2. **Project plan & URS (User Requirement Specification):** To be submitted by Week 3.
3. **UML Class Diagram:** Intermediate and final versions to be submitted between Week 5 and Week 16.
4. **Desktop Application:** Intermediate and final versions, submitted throughout the semester.
5. **Web Application:** Intermediate and final versions, submitted throughout the semester.
6. **Test Plan & Report:** To be created and updated by Week 12 and beyond.
7. **Unit tests:** Test cases for happy and edge cases, delivered progressively throughout the semester.

5. PHASING

Phase 1: Ideation and Planning (Weeks 1-3)

- **Goal: Think of a project idea and establish a roadmap.**
 - **Task 1.1: Complete and submit the Ideation document (Week 1).**
 - **Task 1.2: Create the Project Plan .**
 - **Task 1.3: Submit the Project Plan (Week 3).**

Phase 2: Design and Initial Setup (Weeks 4-6)

- **Goal: Design the system's architecture and set up the project infrastructure.**
 - **Task 2.1: Design the UML Class Diagram.**
 - **Ensure correct relationships (inheritance, aggregation, etc.) and notation.**
 - **Review and refine based on teacher feedback.**
 - **Task 2.2: Begin developing the MSSQL database schema.**
 - **Create tables for users, reservations, and inventory (ski and snowboard equipment).**

- Add primary keys, foreign keys, and relationships.
- Task 2.3: Start the Desktop application.
 - Set up the project in .NET Core Windows Forms.
 - Implement basic CRUD functionality for managing equipment and reservations.
 - Introduce one new entity related to equipment
- Task 2.4: Submit an intermediate version of the UML Class Diagram and Desktop application (Week 5).

Phase 3: Web Application Development (Weeks 4-10)

- Goal: Develop the web application using ASP.NET Core Razor Pages.
 - Task 3.1: Implement user registration, login, and reservation features.
 - Create Razor Pages for user interaction with dynamic content (reservations).
 - Fetch data from the MSSQL database (without ORM or LINQ).
 - Implement input validation and error handling.
 - Task 3.2: Implement advanced sorting, filtering, and search capabilities.
 - Enable users to search for equipment by date, type, and availability.
 - Task 3.3: Review web application implementation with WAD teacher.
 - Task 3.4: Submit an intermediate version of the web application (Week 6).

Phase 4: Integration and Advanced Features (Weeks 11-13)

- Goal: Enhance both applications and integrate advanced features.
 - Task 4.1: Continue development of the Desktop application.
 - Task 4.2: Implement advanced features in the Web application.
 - Task 4.3: Implement unit tests for the web and desktop applications.
 - Write tests for reservation creation, equipment availability, and user management.
 - Task 4.4: Review and refine both applications based on teacher feedback.
 - Task 4.5: Submit intermediate versions of both applications, including UML updates (Week 11).

Phase 5: Testing and Finalization (Weeks 14-16)

- **Goal: Finalize the project, perform testing, and submit final deliverables.**
 - **Task 5.1: Complete the Test Plan and conduct thorough testing.**
 - **Ensure both applications are bug-free and handle errors properly.**
 - **Test happy-flow scenarios and edge cases.**
 - **Task 5.2: Finalize the MSSQL database, ensuring all tables have the required rows (20,000+).**
 - **Include views and aggregated data for statistical purposes.**
 - **Task 5.3: Perform final reviews and refactor code as needed to meet OO principles.**
 - **Task 5.4: Submit the final version of all deliverables before the end of Week 16.**

6. MILESTONES AND DEADLINES

Milestone	Deadline
Ideation document submitted	Week 1
Project plan and URS submitted	Week 3
Intermediate UML & Desktop App submitted	Week 5
Intermediate Web App submitted	Week 6
Final UML & Desktop App submitted	Week 16
Final Web App submitted	Week 16
Final Test Plan and Report submitted	Week 16
Final Database and Statistical Reports submitted	Week 16

7. CONSTRAINTS

- **ASP.NET Core Razor Pages:** For developing the web application.
- **Windows Forms .NET Core:** For building the desktop application.
- **MSSQL:** As the shared database for both applications.
- **GitLab:** For version control and collaboration with teachers.
- **Unit Testing Frameworks:** For testing the web and desktop applications